

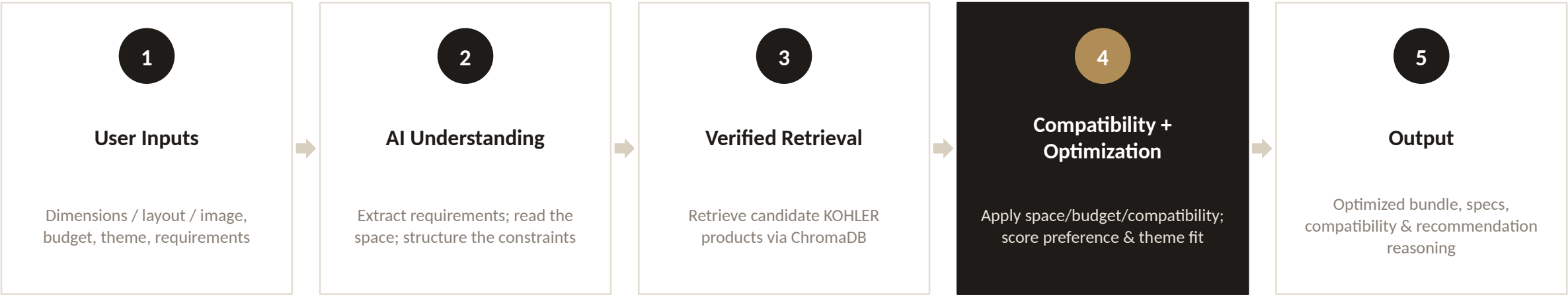
Core Approach

The Problem

Customers must juggle dimensions, layout, product compatibility, budget, function and style at once — product search alone doesn't solve this multi-constraint configuration problem.

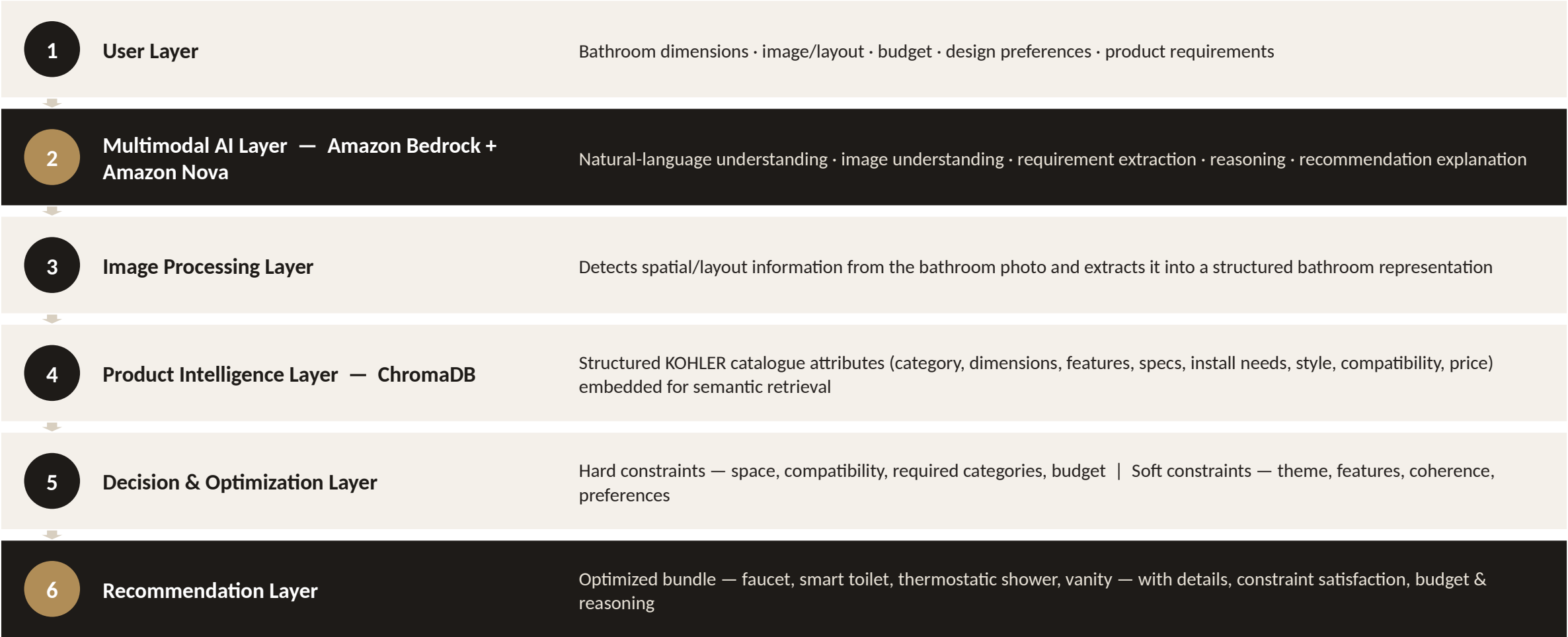
The Objective

An AI design assistant that turns customer requirements and bathroom-space information into a feasible, personalized, budget-aware KOHLER product bundle.



Central concept — AI understanding + verified catalogue retrieval + constraint optimization. Not free-form suggestion — bathroom planning treated as a constrained configuration problem.

System Architecture & End-to-End Flow



Technology Stack & Optimization Engine

Frontend React 19 · Vite 7 · TypeScript · Tailwind CSS · shadcn/ui	Backend Node.js · Express · tRPC · TypeScript · Zod · Drizzle ORM
Data MySQL / TiDB · S3-compatible storage · KOHLER catalogue	AI & Intelligence Amazon Bedrock/Nova (integration-ready) · Python/FastAPI · ChromaDB (integration-ready) · RAG
Core Engines Catalogue retrieval · spatial intelligence · compatibility engine · colour intelligence · lookbook intelligence · constraint & budget optimization	DevOps & Testing Docker · Vitest · Pytest · Git/GitHub

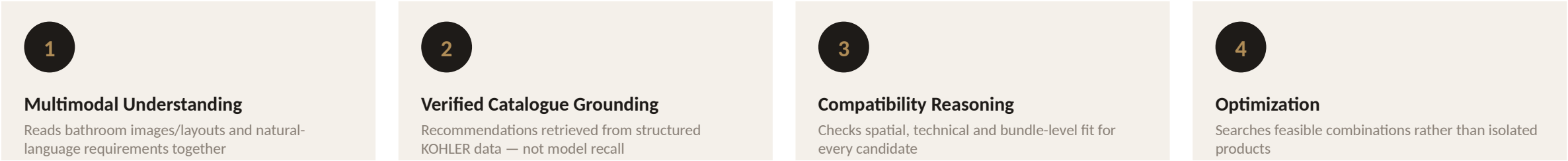
Decision Process

1. Candidate Retrieval
2. Hard-Constraint Filtering
3. Compatibility Filtering
4. Preference Scoring
5. Bundle Optimization
6. Final Recommendation

Optimization strategy — maximize overall bundle suitability (compatibility, space fit, preference & theme match, feature relevance) while penalizing budget deviation and constraint violations, subject to hard constraints.

Feasibility before preference — a visually preferred combination is never recommended if it violates space, budget or compatibility requirements.

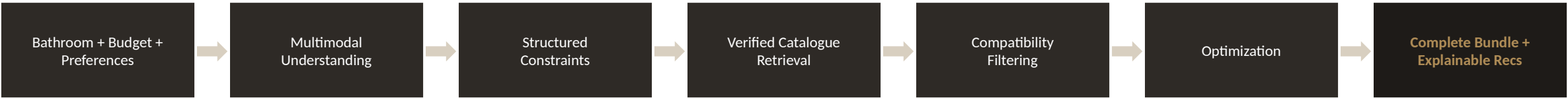
Innovation Pitch & Differentiation



Generic AI Recommendation



KOHLER AI Bathroom Designer



Example — 8 ft × 6 ft bathroom, ₹2,00,000 budget, Japanese Zen theme, needs a smart toilet, faucet, thermostatic shower and vanity → understands the brief, retrieves verified candidates, filters incompatible/space/budget-violating options, optimizes the combination, and outputs a complete bundle with reasoning.